Figure 6.2. Evolutionary algorithms are different from gradient descent-based optimization techniques. With evolutionary strategies, we generate agents and pass the most favorable weights down to the subsequent agents.

Figure 6.3. In an evolutionary algorithm approach to reinforcement learning, agents compete in an environment, and the agents that are more fit (those that generate more rewards) are preferentially copied to produce offspring. After many iterations of this process, only the most fit agents are left.

Figure 6.4. A genetic algorithm optimization of neural networks for reinforcement learning. A population of initial neural networks (the RL agents) is tested in the environment, earning rewards. Each individual agent is labeled by how fit it is, which is based on the rewards earned. Individuals are selected for the next generation based on their fitness; more fit individuals are more likely to be included in the next generation. The selected individuals “mate” and are “mutated” to increase genetic diversity.

Figure 6.5. A string diagram outlining the major steps in a genetic algorithm for evolving a set of random strings toward a target string. We start with a population of random strings, compare each to the target string, and assign a fitness score to each string based on how similar it is to the target string. We then select high-fitness parents to “mate” (or recombine) to produce children, and then we mutate the children to introduce new genetic variance. We repeat the process of selecting parents and producing children until the next generation is full (when it’s the same size as the starting population).

Figure 6.6. This is a plot of average population fitness over the generations. The average population fitness increases fairly monotonically and then plateaus, which looks promising. If the plot was very jagged, the mutation rate might be too high or the population size too low. If the plot converged too quickly, the mutation rate might be too low.

Figure 6.7. We will use the CartPole environment to test our agent. The agent is rewarded by keeping the pole upright, and it can move the cart left or right.